University of Freiburg

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Bachelor Project Report

Localization of a point in a two dimensional triangular space through direct and reflected signals

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Abstract

This project investigates the framework of Indoor Localization through Direct And Reflected Signals (ILDARS) within a two-dimensional, triangular space. The setup consists of a single sender and a single receiver enclosed by three reflecting walls. The primary objective is to analyse the geometric and acoustic properties of this specific configuration to determine a robust solution. To achieve this, we developed a custom Python simulator to model ray-propagation dynamics and analyse the patterns formed by image sources across different reflection orders.

By exploiting these patterns and the distance-preserving properties of reflections (isometries), we formulate a closed-form quadratic equation as a solution. Ultimately, this approach computes both the shape of the room and the position of the sender, provided there is sufficient signal coverage.

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Part I

Setup

Starting we should first understand the given setup in more detail, so we can be consistent through out the entire report. It's important to mention, that the analysis only covers triangles, so three distinct walls, where none are parallel or identical to each other. Throughout this report, the environment is assumed to be in a general position, meaning no degenerate geometric configurations exist among the walls, the sender, or the receiver. Furthermore, both the sender and the receiver are strictly constrained to the interior of the triangular room. The reconstruction framework relies on two primary acoustic principles:

- Time Difference of Arrival (TDOA), which contains the time difference of each Signal, relative to the first received signal. This means that the first signal received arrives at time 0. All the other signals $j \in \mathbb{N}$ have a TDoA of δ_j.
- Angle of Arrival (AoA) to each signal j we also get its angle of arrival, relative to the receiver. This will always be in vector representation, so for each signal we receive $(\vec{v}_j, \delta_j)$.

It is assumed that we always receive a direct signal I_0 with $(\vec{v}_0, \delta_0)$. But $\delta_0 = 0$, because it is always the first signal, which arrives. This is guaranteed by the **triangle inequality**.

1 Visualization

The proposed geometric model is illustrated in Figure 1. This figure demonstrates a possible spatial configuration containing the three walls (blue), the sender (blue), and

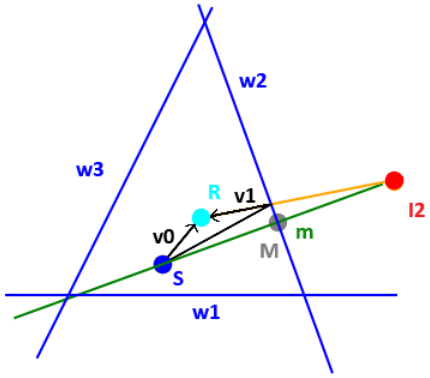


Figure 1: Geometric configuration of the triangular room setup.

the receiver (cyan). The acoustic paths, illustrated as black rays, trace the sound events from their origin at the sender through their reflections off the respective boundaries.

1.1 Notation

Having established the core elements in the setup, we now want to formally declare them (Table 1).

Object	Notation	Description
Wall	$\vec{w}_k : \ k \in \{1,2,3\}$	Wall which reflects sound or mirror which reflects a ray.
Sender	S	Source of all the sound events/rays. Some 2D point inside the triangle.
Receiver	R	The observers Point of view, receives all the rays (the **AoA** vectors $\vec{v}_j$). Also R is always $(0,0)^{-1}$.
Signals	$\vec{v}_j : \ j \in \mathrm{N}$	All the **AoA**'s of the received signals by the receiver (not received or signals received in too high order of reflection are neglected).
Image points	$I_l : \ l \in \{1,2,3\}^o$	This is where the receiver sees a signal, solely depending on the tuple, which he receives. The notation is always I for image point and then the walls in the exact order the signal got reflected by. The only exception being the direct signal I_0. And o represents the order of reflection, because each reflections has to happen at some wall. An example would be I_{132312}, but what can never happen is some number being twice in a row obviously.
Middle point	M	The point, where the orthogonal vector of the wall intersects the sender or the previous image point. Mainly serves a computational purpose, to compute the next image point.
Orthogonal line	m	A line with the direction of the orthogonal wall vector of the respective wall. Line between previous image point and new one. The distance between those two is exactly the distance between each of them to M twice, hence M is the middle point.

Table 1: Project notations

2 Formulae

Now we look at all the formulas representing the different objects in our setup. First we start with the signals. As we already know, each signal is received as a tuple:

$$I_l = (\vec{v}_j, \delta_j)$$

The l is the reflection order starting with 0. This only occurs, when it is the direct signal. Because we work with **TDOA** $\delta_0 = 0$, but we know that sound moves at approximately $343m/s$, so it does always need more then 0 time to arrive at the receiver. So we introduce the variable p and say that $\delta_0 = p$, which is the unknown propagation distance of the direct signals. All other signals with index $j > 0$ have a propagation distance of exactly $p + \delta_j$. And this is exactly the way, how each signal is reconstructed from the receivers point of view:

$$p \cdot \vec{v}_j + \delta_j \cdot \vec{v}_j = I$$

This is just some arbitrary point in the 2D space, but with the relative distance to the receiver.

3 Oracle World

The solution, which will be provided only works in the **Oracle World**, where we know for each received signal I_l, the exact l. So in other words we know this signal's order of reflection and also the order of the reflecting walls. This may seem very unrealistic, but it actually is not that sophisticated, because for the solution we only need a signal of order two, hence $l \in \{1, 2, 3\}^2$.

Part II

Simulator

This part introduces the *Python* simulator build for this project. It's goal is to simulate any realistic triangle, with one sender and receiver inside. It uses the Python library **matplotlib.pyplot** to plot the results of the simulations. To simulate an acoustic **source**, the sender emits a discrete number of signals uniformly distributed across all horizontal angles from 0 to 2π radians. The angular resolution is controlled by a parameter N. For each discrete step $x \in \{0, \ldots, N-1\}$, the initial propagation direction vector is computed using trigonometric functions to ensure uniform angular spacing. Then all the rays get traced until they either reach the **receiver** or reach the maximum **order of reflection**. This way we have information about all the relevant rays. In addition we only show the rays, which are relevant, so the triangle does not get filled with rays, which do not contribute to the simulation result. Then we can do things like display these rays, show the image points, recognise relations or find patterns.
One **Remark**, because we are talking about code here, which is commented itself, i don't elaborate on the source code too much. Also not on the calculations, too compute things like the image point, because it is trivial geometry, and can be researched easily.

1 Motivation

The motivation behind the simulator was to find a pattern in the image point, which is somewhat simple to distinguish, such that we can use these patterns to find a general solution for the three wall problem. This originated from Marcel Schuhmacher's bachelor thesis, where he found together with Prof. Dr. Schindelhauer a pattern to

be used. [2] This was a circle with it's centre being the intersection of the two walls, which were present in their setup.

2 Architecture

The code base uses **Object oriented programming**. The primary classes are:

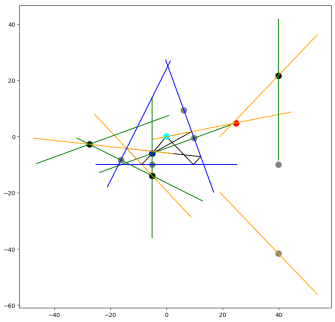
Name	Description	Tasks
Line	Represents a 2D Line, so a direction vector and an anchor vector. So it can compute things like checking whether a point is on the line, finding an intersecting point with another line, finding the perpendicular distance point and computing the image point.	This is basically the main computation of the entire setup, because here we provide all the methods necessary for ray tracing.
Ray	This is just a ray in the 2D space inside the triangle. So basically a collection of vectors. It uses the line class, because the initial vectors are undefined, these are first variable length lines, until we found the intersections.	The only task it has is to expand a ray, so we get a starting vector from the source and we want to trace it *max_order* times or until we hit the source. If nothing is hit and no more reflections are allowed the ray stays empty, so we have an end product of only the rays, which are relevant.
Engine	Runs all the calculations in the correct order, so we can simulate our setup. Doesn't use much code, because it is embedded in the classes themselves. Also generates the final product, using matplotlib.	Generates walls, creates initial starting rays. Plots all the successful rays and image points etc. if desired.

Table 2: Main Classes

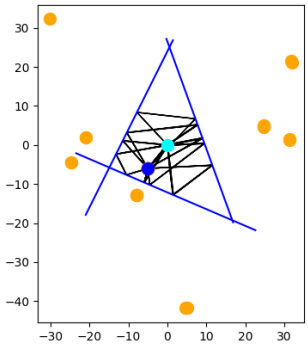
Every class in has a method *plot()*, which plots the object. This way we can just plot all objects that we want to show in the matplotlib end product with *obj.plot()*.

3 Results

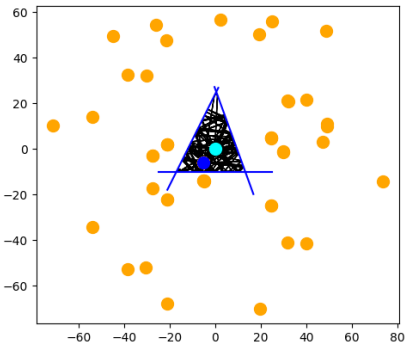
The results are all plots, but of many different kinds. This comes from the fact, that you can show certain toggle certain things, or not toggle them.



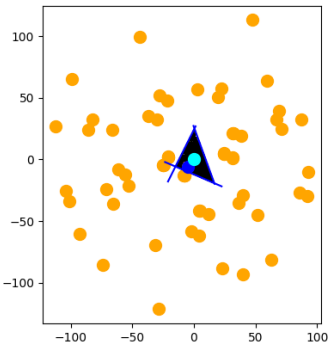
(a)



(b)



(c)



(d)

Figure 2: Simulator Outputs

3.1 Internal computation

We can observe in Figure 2a, how each reflection is computed internally, using the image method. First the perpendicular point to the reflecting line is found. Second the distance between that point and the perpendicular point is computed. Then a new vector is created from the intersection point and the newly computed mirrored point. The resulting vector is then the reflected rays direction.

3.2 Low order of reflection

In Figure 2b we can see the output of a triangle where only order two reflection is allowed. We can see, that the rays get reflected at most twice, because of this upper bound. Another important thing to observe is, that some points are *blurry*, this comes from the receiver having a radius of 0.125, such that we receive all the crucial information and don't loose points to python floating point limit.

3.3 Medium order of reflection

In Figure 2c we allow 4-th order of reflection, and we can already observe, that the points distance form the triangle gets way larger. We can also start to recognise some patterns Part III. The inside of the triangle gets really dark, because more rays actually reach the receiver.

3.4 High order of reflection

With a maximum reflection order of 7 in Figure 2d, we can already observe, how many rays get to the receiver, because the triangle itself is completely blacked out. Even more patterns are visible and the distance to the triangle obviously increases.

Part III

Analysing the simulation results

In this part we want to use the results and analyse their pattern, to find a solution, relying on such pattern. For this to work we need patterns, which are always present, no matter the circumstances are. To actually analyse patterns it seems to be easier to take higher order of reflections.

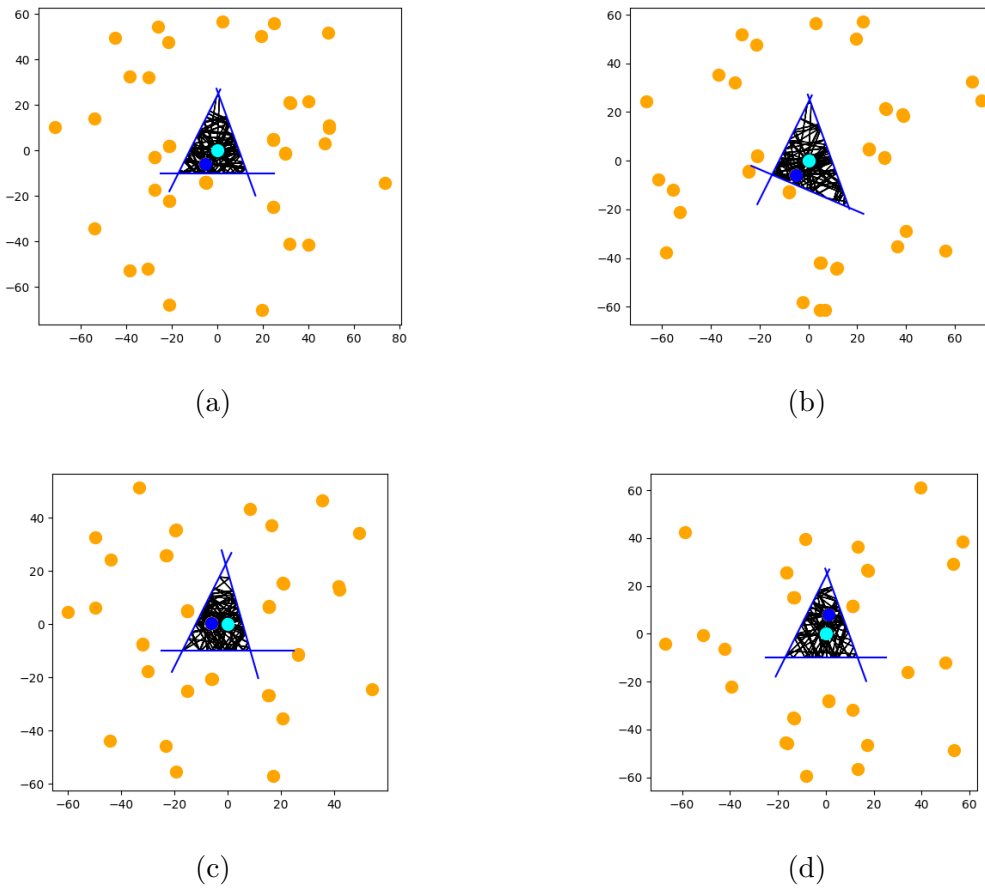
1 Reoccurring patterns in different setups

For proper analysis we fix some parameters, first we fix the order of reflection to 4 and the number of signals to 1440 (See Figure 3). Then we have the four following results, with different triangles and sender positions. The focus will be shifted towards clearly characterizing all the the patterns, which we can observe in every single setup and also to those which seem more unique, to one or two setups.

1.1 Similarities

A major similarity across all setups is the presence of two distinct types of circular structures. The first type consists of primary circles centred directly on the physical vertices of the triangle, which are generated by localized, alternating reflections between two adjacent walls [2].

As shown in Figure 3b, these vertex-anchored circles appear across various setups, though not always at the same vertices. Their visibility depends on the sender and receiver positions, as certain geometric configurations make specific reflection paths harder to form. These circles arise exclusively from alternating reflections between two walls. For instance, considering a vertex formed by walls w_i and w_j, the corre-



(a) (b) (c) (d)

Figure 3: Comparison of all scenarios.

sponding circle contains only image sources whose reflection history solely consists of w_i and w_j.

The second type consists of higher-order circles whose centres lie outside the physical boundaries of the room. These occur when reflection sequences traverse all three walls, projecting virtual centres deeper into the image plane.

A clear example, visible in all plots of Figure 3, is the circle located below the baseline. Its distance to the triangle varies depending on the sender's position relative to the

baseline. This circle appears particularly prominent in Figures 3a, 3c, and 3d because the baseline reflection plays a dominant role, effectively mirroring the circle above the baseline across that boundary.

This effect is not restricted to the baseline; it is simply more noticeable in these scenarios. Fundamentally, all circles that are not directly anchored to a physical vertex are created by mirroring vertex-anchored circles across the boundary walls.

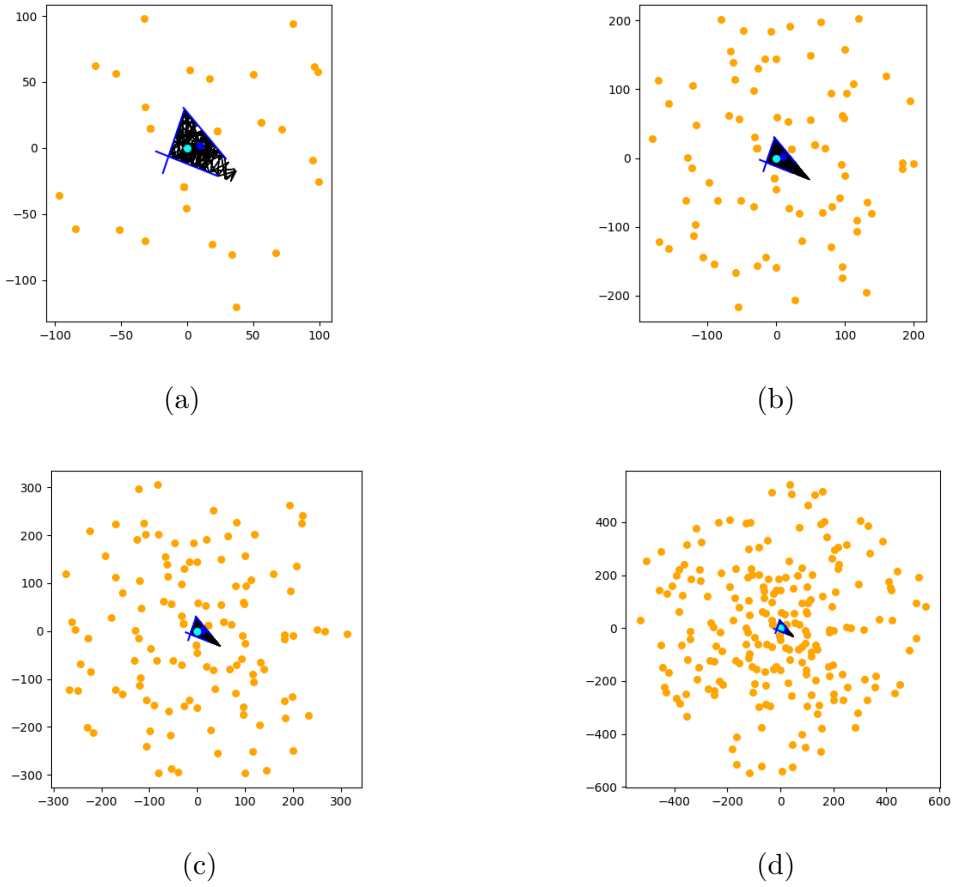
1.2 Differences

A key difference between the simulated scenarios is the appearance of uneven point densities and structural gaps, which represent acoustic shadow zones. Depending on the interior angles of the triangle, reflections fail to hit the next wall, causing rays to escape the room. In the plots, this leads to asymmetric point distributions. While some setups create dense circular clusters (Figure 3b), others show empty areas where geometric constraints prevent reflection paths from reaching the receiver (Figure 3d). For example, when the sender or receiver is close to a tight corner, the narrow angle limits valid ray paths, leaving visible gaps in the outer rings (Figure 3a).

The relative positions of the sender and receiver also change how the patterns look. When the receiver is placed directly above the sender along a primary reflection line, the symmetry produces clear *double circles* with even spacing, as seen in Figures 3a and 3b. Shifting the receiver away from these lines of symmetry breaks this structure, spreading the points into wider, less defined alignments. Additionally, if the distance between the sender and receiver is small, the relative time delays (δ_j) remain small, creating tight point clusters (Figures 3a–3c). As the sender-receiver distance grows, the larger differences in delay expand the geometric spread, making the circular patterns appear wider and more scattered (Figure 3d). This shows how closely the visual pattern of image sources depends on the room's shape and setup.

2 Pattern Behaviour with increasing Reflection Orders

Allowing higher orders of reflection introduces a noticeable evolution in the spatial distribution of virtual image points. As the maximum reflection depth increases, certain structural patterns become more pronounced, while others degrade into isolated scattering.



(a) (b)

(c) (d)

Figure 4: Comparison of different orders of reflection

2.1 Persistent Geometric Structures: Circles and Arcs

A primary observation as reflection orders increase is the continuous emergence and refinement of circular patterns and arc segments. Because every successive reflection across a pair of non-parallel walls acts mathematically as a rotational isometry around their intersection vertex, the generated image sources inherently follow circular trajectories. Consequently, no rigid transformations other than circular rotations are generated by these multi-bounce pathways. Higher reflection depths do not deform these fundamental curves; rather, they populate more points along the perimeter of these concentric circles and semi-circular arcs, making their radii and centres more distinct.

2.2 Degradation and Scattering Effects

Conversely, increasing the reflection order also introduces complex spatial dispersion that can obscure recognizable structures. While low-order reflections yield sharp, tightly bound geometric features, higher-order reflections result in a lower recognizable point density per individual pattern. A significant portion of higher-order image points scatter in isolation across the coordinate space due to complex, non-repeating sequence paths across all three walls. Furthermore, as the maximum delay increases, the global layout of image points expands outward radially, placing the physical triangular room at the center of a large, dartboard-like target structure characterized by concentric rings and radially scattered points.

3 Takeaway

The visual analysis of the simulation results demonstrates that structured patterns—particularly circular trajectories anchored to room geometry—are persistently present across all scenarios. However, relying purely on visual or heuristic pattern

recognition to solve the room reconstruction problem presents severe challenges. As the order of reflection increases, the sheer abundance and overlapping nature of these patterns, combined with isolated scattering points, creates high structural ambiguity. This highlights the necessity for a deterministic algebraic solution, rather than heuristic pattern matching, to uniquely resolve the room geometry and source position.

Part IV

Solution to the problem

Now we want to approach the problem through the oracle setup, where we use our I_l notation. Because there is a more natural way, which is just equations. The intuitive idea would be to just overload an equation with information, so with different I_l values, what we receive. But this would result in some equations of order 7 or higher possibly, which there is no known solution for. So we need a little trick.

1 What is necessary?

So the trick consists of two things, the first thing would be, that mirroring an object does not change it's size. We need to produce different things, for the solution to work, we need the direct signal I_0, three signals, which have been reflected once I_1, I_2, I_3, each from different walls and one arbitrary signal with reflection of order two $I_{ij} : i, j \in \{1, 2, 3\}$.
It would be easier to know, how the order two signal got reflected, but we can also try out and check later. This is not inefficient, because there are only 6 different possibilities.

1.1 Isometry

The law, which we want to use is, that the distance between the **Sender** and some $I_j : j \in \{1, 2, 3\}$, is always the same as the distance between both of these points reflected at some wall $w_k : k \in \{1, 2, 3\} \wedge k \neq j$.
This is because both these points are just mirrored on the wall w_k.

So we can use the resulting equation [1]

$$\|S - I_j\|^2 = \|I_k - I_{jk}\|^2$$

This comes from the Isometry
The equation can be used to find two possible values for p, where one may be ruled out by being negative.

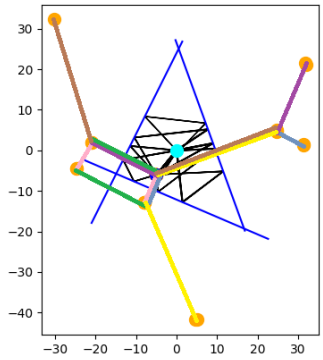


Figure 5: Equal distance point pairs

In Figure 5 the said equation is visualized. We can see that each point pair, where both signals get reflected by the same wall still have the same distance.
Each line pair with the same colour represents one of these equalities occurring in the specific setup.
We can also see, that some equalities, like *brown* and *yellow*, *green* and *violet* and *pink* and *grey*, each of them are also equal.
this comes from the shared line being able to be mirrored on both of the other present mirrors. Even though we can see this happening for all our point pairs in this case, this must not hold always, because not all signals are present in every setup.

2 Solution in Oracle case

Now we want to clearly state the formulae, which we need to compute p in our oracle setup.
The equation, which we know of would be

$$\|S - I_j\|^2 = \|I_k - I_{jk}\|^2 \tag{1}$$

but we can also change S to I_0, because this is just the view, which the receiver has of the source, via direct signal.

$$\|I_0 - I_j\|^2 = \|I_k - I_{jk}\|^2 \tag{2}$$

Let's change it to the vector representation and also add δ_0, even though it is always 0. This way we have a more general formulation.

$$\|\vec{v}_0 p + \vec{v}_0 \delta_0 - (\vec{v}_j\, p + \vec{v}_j\, \delta_j)\|^2 = \|\vec{v}_k\, p + \vec{v}_k\, \delta_k - (\vec{v}_{jk}\, p + \vec{v}_{jk}\, \delta_{jk})\|^2, \tag{3}$$

We transform this representation into an indexed one.

$$\left\| \begin{pmatrix} v_{0,1}p + v_{0,1}\delta_0 - v_{j,1}p - v_{j,1}\delta_j \\ v_{0,2}p + v_{0,2}\delta_0 - v_{j,2}p - v_{j,2}\delta_j \end{pmatrix} \right\|^2 = \left\| \begin{pmatrix} v_{k,1}p + v_{k,1}\delta_k - v_{jk,1}p - v_{jk,1}\delta_{jk} \\ v_{k,2}p + v_{k,2}\delta_k - v_{jk,2}p - v_{jk,2}\delta_{jk} \end{pmatrix} \right\|^2 \tag{4}$$

Now we can apply the distance of a vector formula.

$$(v_{0,1}p + v_{0,1}\delta_0 - v_{j,1}p - v_{j,1}\delta_j)^2 + (v_{0,2}p + v_{0,2}\delta_0 - v_{j,2}p - v_{j,2}\delta_j)^2 = \tag{5}$$

$$(v_{k,1}p + v_{k,1}\delta_k - v_{jk,1}p - v_{jk,1}\delta_{jk})^2 + (v_{k,2}p + v_{k,2}\delta_k - v_{jk,2}p - v_{jk,2}\delta_{jk})^2 \tag{6}$$

The next step is to combine all the factors of p.

$$((v_{0,1} - v_{j,1})p + v_{0,1}\delta_0 - v_{j,1}\delta_j)^2 + ((v_{0,2} - v_{j,2})p + v_{0,2}\delta_0 - v_{j,2}\delta_j)^2 = \tag{7}$$

$$((v_{k,1} - v_{jk,1})p + v_{k,1}\delta_k - v_{jk,1}\delta_{jk})^2 + ((v_{k,2} - v_{jk,2})p + v_{k,2}\delta_k - v_{jk,2}\delta_{jk})^2 \tag{8}$$

Now we apply the first two binomial formulae twice on each outer bracket.

$$(v_{0,1} - v_{j,1})^2p^2 - 2(v_{0,1} - v_{j,1})(v_{0,1}\delta_0 - v_{j,1}\delta_j)p + v_{0,1}^2\delta_0^2 - 2v_{0,1}\delta_0 v_{j,1}\delta_j + v_{j,1}^2\delta_j^2 + \tag{9}$$

$$(v_{0,2} - v_{j,2})^2p^2 - 2(v_{0,2} - v_{j,2})(v_{0,2}\delta_0 - v_{j,2}\delta_j)p + v_{0,2}^2\delta_0^2 - 2v_{0,2}\delta_0 v_{j,2}\delta_j + v_{j,2}^2\delta_j^2 = \tag{10}$$

$$(v_{k,1} - v_{jk,1})^2p^2 - 2(v_{k,1} - v_{jk,1})(v_{k,1}\delta_k - v_{jk,1}\delta_{jk})p + v_{k,1}^2\delta_k^2 - 2v_{k,1}\delta_k v_{jk,1}\delta_{jk} + v_{jk,1}^2\delta_{jk}^2 + \tag{11}$$

$$(v_{k,2} - v_{jk,2})^2p^2 - 2(v_{k,2} - v_{jk,2})(v_{k,2}\delta_k - v_{jk,2}\delta_{jk})p + v_{k,2}^2\delta_k^2 - 2v_{k,2}\delta_k v_{jk,2}\delta_{jk} + v_{jk,2}^2\delta_{jk}^2 \tag{12}$$

Gather all the p^2, p and the *constants*.

$$(v_{0,1} - v_{j,1})^2p^2 + (v_{0,2} - v_{j,2})^2p^2 - (v_{k,1} - v_{jk,1})^2p^2 - (v_{k,2} - v_{jk,2})^2p^2 \tag{13}$$

$$-2(v_{0,1} - v_{j,1})(v_{0,1}\delta_0 - v_{j,1}\delta_j)p - 2(v_{0,2} - v_{j,2})(v_{0,2}\delta_0 - v_{j,2}\delta_j)p \tag{14}$$

$$+2(v_{k,1} - v_{jk,1})(v_{k,1}\delta_k - v_{jk,1}\delta_{jk})p + 2(v_{k,2} - v_{jk,2})(v_{k,2}\delta_k - v_{jk,2}\delta_{jk})p \tag{15}$$

$$+v_{0,1}^2\delta_0^2 - 2v_{0,1}\delta_0 v_{j,1}\delta_j + v_{j,1}^2\delta_j^2 + v_{0,2}^2\delta_0^2 - 2v_{0,2}\delta_0 v_{j,2}\delta_j + v_{j,2}^2\delta_j^2 + v_{k,1}^2\delta_k^2 \tag{16}$$

$$-2v_{k,1}\delta_k v_{jk,1}\delta_{jk} + v_{jk,1}^2\delta_{jk}^2 + v_{k,2}^2\delta_k^2 - 2v_{k,2}\delta_k v_{jk,2}\delta_{jk} + v_{jk,2}^2\delta_{jk}^2 = 0 \tag{17}$$

Again combine all the factors.

$$((v_{0,1}-v_{j,1})^2+(v_{0,2}-v_{j,2})^2-(v_{k,1}-v_{jk,1})^2-(v_{k,2}-v_{jk,2})^2)p^2 \tag{18}$$

$$+(-2(v_{0,1}-v_{j,1})(v_{0,1}\delta_0-v_{j,1}\delta_j)-2(v_{0,2}-v_{j,2})(v_{0,2}\delta_0-v_{j,2}\delta_j) \tag{19}$$

$$+2(v_{k,1}-v_{jk,1})(v_{k,1}\delta_k-v_{jk,1}\delta_{jk})+2(v_{k,2}-v_{jk,2})(v_{k,2}\delta_k-v_{jk,2}\delta_{jk}))p \tag{20}$$

$$+v_{0,1}^2\delta_0^2-2v_{0,1}\delta_0 v_{j,1}\delta_j+v_{j,1}^2\delta_j^2+v_{0,2}^2\delta_0^2-2v_{0,2}\delta_0 v_{j,2}\delta_j+v_{j,2}^2\delta_j^2+v_{k,1}^2\delta_k^2 \tag{21}$$

$$-2v_{k,1}\delta_k v_{jk,1}\delta_{jk}+v_{jk,1}^2\delta_{jk}^2+v_{k,2}^2\delta_k^2-2v_{k,2}\delta_k v_{jk,2}\delta_{jk}+v_{jk,2}^2\delta_{jk}^2=0 \tag{22}$$

And finally we have all the constants $A, B, C \in \mathbb{R}$, to solve the quadratic equation through the quadratic formula.

$$A=(v_{0,1}-v_{j,1})^2+(v_{0,2}-v_{j,2})^2-(v_{k,1}-v_{jk,1})^2-(v_{k,2}-v_{jk,2})^2 \tag{23}$$

$$B=-2(v_{0,1}-v_{j,1})(v_{0,1}\delta_0-v_{j,1}\delta_j)-2(v_{0,2}-v_{j,2})(v_{0,2}\delta_0-v_{j,2}\delta_j)+ \tag{24}$$

$$2(v_{k,1}-v_{jk,1})(v_{k,1}\delta_k-v_{jk,1}\delta_{jk})+2(v_{k,2}-v_{jk,2})(v_{k,2}\delta_k-v_{jk,2}\delta_{jk}) \tag{25}$$

$$C=+v_{0,1}^2\delta_0^2-2v_{0,1}\delta_0 v_{j,1}\delta_j+v_{j,1}^2\delta_j^2+v_{0,2}^2\delta_0^2-2v_{0,2}\delta_0 v_{j,2}\delta_j \tag{26}$$

$$+v_{j,2}^2\delta_j^2+v_{k,1}^2\delta_k^2-2v_{k,1}\delta_k v_{jk,1}\delta_{jk}+v_{jk,1}^2\delta_{jk}^2+v_{k,2}^2\delta_k^2-2v_{k,2}\delta_k v_{jk,2}\delta_{jk}+v_{jk,2}^2\delta_{jk}^2 \tag{27}$$

After applying the quadratic formula and perhaps ruling out one of the values for p, we can simply reconstruct the triangle.
The Sender will be at $p \cdot \vec{v}_0$. The walls are a little more complicated, but we can just reverse the image point method to compute the walls.
For each wall w_k we compute the respective M_k with

$$M_k=\left(\frac{I_{0,1}+I_{k,1}}{2},\frac{I_{0,2}+I_{k,2}}{2}\right)$$

Then we compute each wall's orthogonal vector $\vec{w}_k$, with $\vec{w}_k=\overrightarrow{M_k I_k}$ and use M_k as the anchor point for each w_k.

This way we can reconstruct the entire room and the senders position.

3 Extension of the solution

Now we have to look at where this solution comes from. As we know, from its derivation it is just derived by defining it as a mirroring of I_0 and I_j at the mirror w_k, s.t. we get I_k and I_{jk}.
So the only real constant the we still have is the 0, which we can eliminate and set it to i, because it would still be just a mirroring with I_i and I_j to I_{ik} and I_{jk}.
So when looking at the reflection tree, fixing any two points, with the most recent reflection not being k, then going one more reflection down in the tree in the direction of k, creates a mirroring of the images, s.t. our condition applies.
We achieve the following general equation for any points, which the stated conditions hold for.

$$\|I_i - I_j\|^2 = \|I_{ik} - I_{jk}\|^2$$

It is important to know, that I_i and I_j can be any pair of points in the reflection tree, which don't violate our conditions, so in this case $i \in \{1,2,3\}^o$ and $j \in \{1,2,3\}^o$.
A very important remark is, that extending the solution like this, so de anchoring it from our sender, hence generalize it more makes the reconstruction of the geometry way harder and a completely new, different problem.

Part V

Conclusion

In this project, we investigated the problem of Indoor Localization through Direct And Reflected Signals (ILDARS) within a two-dimensional triangular enclosure. By

analyzing the acoustic ray propagation between a single sender and receiver bounded by three non-parallel reflecting walls, we established both an empirical visualization framework and a deterministic mathematical solution for room and source reconstruction.

To study the underlying ray-tracing dynamics, we developed an object-oriented Python simulation environment. The simulator enabled a structured analysis of virtual image source configurations across various reflection orders[cite: 2]. Visualizing these image point distributions revealed persistent geometric structures, most notably concentric circular trajectories and arc patterns anchored to room vertices[cite: 2]. However, our investigation also highlighted that relying purely on visual or heuristic pattern recognition becomes increasingly infeasible at higher reflection orders due to severe structural overlap, acoustic shadow zones, and point-cloud scattering[cite: 2].

To overcome these visual ambiguities, we presented a closed-form algebraic solution operating within an oracle model[cite: 2]. By leveraging the isometric, distance-preserving properties of specular reflections[cite: 2], we formulated a system of equations that reduces the determination of the unknown propagation distance p to a standard quadratic equation[cite: 2]. This approach avoids high-degree polynomial root-finding and allows for the direct reconstruction of both the sender's spatial position and the room's boundary walls using low-order signal reflections[cite: 2]. Furthermore, we demonstrated that this isometric relationship can be generalized across arbitrary nodes in the reflection tree[cite: 2].

While unanchoring the solution from direct first-order reflections expands the theoretical versatility of the framework, it shifts the secondary challenge toward complex inverse geometric reconstruction[cite: 2]. Future work could explore integrating robust time-of-arrival ellipse intersections and backward ray-tracing algorithms to resolve room geometry when primary reflection signals are unavailable[cite: 2]. Overall,

the findings of this project demonstrate that combining isometric geometric principles with TDoA and AoA acoustic measurements provides a mathematically sound and effective foundation for two-dimensional indoor localization[cite: 2].

Declaration on the Use of Artificial Intelligence

I hereby declare that I used AI-based language models as an tool during the preparation of this report. Their use was explicitly limited to the correction of grammar and spelling as well as the stylistic improvement of sentence formulations. The conceptual development, mathematical and algorithmic derivations, and scientific argumentation were carried out entirely and independently by me. The use of this tool was discussed and agreed with the first examiner.

_____ _____

Place, Date Signature

Declaration

I hereby declare that I am the sole author and composer of my report and that no other sources or learning aids, other than those listed, have been used. Furthermore, I declare that I have acknowledged the work of others by providing detailed references of said work.
I hereby also declare that my Thesis has not been prepared for another examination or assignment, either wholly or excerpts thereof.

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Place, Date Signature